

Hoon Y.

SUMMARY: Senior ML leader building and shipping **production AI/ML systems** across federal and commercial domains. Proven track record **defining end-to-end ML strategy**, **leading ML teams** and turning complex data into **models that materially move revenue**, fraud/risk, **reduce costs**, and **enhance customer experience** metrics.

Google AI Innovation Award at Brandtech and current doctoral **researcher** in AI.

SELECTED IMPACT HIGHLIGHTS

- **\$200M+ business value created** for USCIS by architecting a statistical measurement platform for large-scale production performance evaluation.
- **\$19–\$23M/year in DHS adjudicator cost savings** by leading development of graph-based false match detection model while using 99% less data and no personal identifiable information.
- **20%–65% reduction in media waste** through forecasting, propensity, and budget-optimization models for digital advertising.
- **6–9× analyst efficiency** from graph-based anomaly-detection pipelines in high-sensitivity security environments.
- **90%+ reduction in manual review** via anomaly-detection and data-quality models for Census 2020.
- **Google AI Innovation Award** for a multimodal CV+NLP creative-understanding platform for Ad-tech / marketing technology.
- **12× more knowledge access** with high-guardrail RAG/LLM systems for Department of Veterans Affairs medical claims processing for enterprise health record workflows.

EXPERIENCE

Senior AI/ML Specialist – Principal Architect, Identity & Measurement

MetroStar – Department of Homeland Security (USCIS) | 2025 – Present | Virginia /

Remote

- Own ML strategy for identity and case-management workflows, defining a multi-year roadmap for entity resolution, risk scoring, and measurement across programs.
- Successfully deployed a statistical measurement system that effectively replaces a manual effort worth \$9.3 million in value for the initial study and scaled higher for approx \$200 million a year in production measurement capacity.

- Lead the design of graph-based entity-resolution models using achieving equivalent match rate with ~99% less labeled data and no direct PII.
- Partner with program leaders to align ML capabilities with mission-critical outcomes.

Head of AI / Senior AI & ML Strategy Consultant – Veterans Affairs Contract

Tojj LLC – AI/ML Consulting | 2023 – 2024 | Remote

- Led the design and deployment of RAG and LLM-powered products that convert siloed text, logs, and knowledge bases into reusable features and insight services, unlocking ~12× more accessible content with ~98% KPI alignment on safety/quality.
- Defining end-to-end ML and GenAI strategy for document understanding, content access, and knowledge-assistant use cases.
- Evaluate and select platforms and tooling (Vertex AI, AWS, Databricks, vector DBs) and advise on in-house vs. third-party investments for search, retrieval, and orchestration.
- Influence senior stakeholders on AI roadmap, risk/benefit trade-offs, and scaling patterns for future ML applications.

Director of Data Science – Multimodal Creative & Ads ML

Brandtech Digital | 2021 – 2024 | Reston, VA

- Owned ML strategy for creative understanding and media efficiency, spanning model design, experimentation, deployment, and continuous improvement.
- Led development of multimodal CV+NLP and GenAI systems that interpreted ad creative content with ~98% accuracy, winning the **Google AI Innovation Award** and directly improving promotion decisions and brand safety.
- Delivered propensity, churn, forecasting, and budget-optimization models that reduced media waste by 20–65% and improved ROAS across major clients.
- Prototyped privacy-preserving attribution pipelines using differential privacy and information-theoretic designs, aligning measurement with platform and regulatory constraints.
- Partnered with senior product and commercial leaders to translate model outputs into client-facing products, dashboards, and optimization levers.

AI/ML Specialist – Network & Behavioral Analytics (Top-Secret)

Mobius Consulting – Department of Defense | 2019 – 2021 | Virginia

- Led ML programs for authorship identification, behavioral profiling, and threat analytics over high-volume communication data.
- Designed and shipped anomaly-detection and graph-intelligence pipelines that replaced manual investigations and delivered **6–9× analyst efficiency gains**.

Analytics Modeling Manager / Senior Data Scientist – Data Quality & Risk

Accenture Federal Services – U.S. Census Bureau | 2017 – 2019 | Greenbelt, MD

- Led anomaly-detection and risk-scoring initiatives for Census 2020, working on massive person/address datasets.
- Built statistical models and indicators that reduced manual review work by **90%+**, effectively turning internal data into reusable risk, quality, and coverage scores.

Lead Engineer – Earth Sciences Data & Analytics

SSAI – NASA Goddard Space Flight Center | 2000 – 2016 | Greenbelt, MD

- Developed time-series forecasting and anomaly-detection pipelines over satellite and climate data to support mission planning and operational decision-making.
- Drove adoption of advanced analytics and early ML techniques, mentoring engineers and analysts on modeling best practices.

EDUCATION

Doctoral Candidate, Artificial Intelligence (Expected 2026)

George Washington University – School of Engineering & Applied Science

Research focus: self-supervised transformer–graph methods for large-scale entity resolution and behavioral modeling.

M.S., Physics – University of Maryland, Baltimore County

B.S., Physics – Purdue University